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SUMMARY

Mechanical Engineer with over three years of experience in mechatronics, manufacturing, and medical device development. Currently serving as a Haptics Engineering Research Assistant and Manufacturing Engineer Fellow, with responsibilities including developing mechatronic systems, designing and testing medical devices, and integrating sensing into manufacturing equipment. Additional experience includes yacht manufacturing and computational fluid dynamics (CFD), with applications in incompressible and compressible flow as well as convective heat transfer. Former Division I baseball athlete and recipient of multiple engineering scholarships.

EXPERIENCE

Haptics Engineering Research Assistant – CHROME Lab, Saint Louis University Oct 2021 - Present

Wearable Haptic Integrated Mindfulness System (WHIMS)- 1st place SLU Sigma XI Poster Presentation

– Designing and testing of novel wearable haptic systems to determine optimal sensation for guided breathing practices
– Developing mechanoelectrical systems to optimize user experience, achieving >90% haptic localization rates
– Integrating actuation and sensors into wearable designs for mechanical control of haptic systems

Blind Ice Hockey Enclosure

– Fabrication and testing of a 3D printed protective device to improve puck durability for the visually impaired
– Designing and prototyping electromechanical noise emitters leveraging accelerometers and radio communication

Lighthouse for the Blind Fellow – St. Louis Lighthouse for the Blind June 2024 – Present

– Designing accessible manufacturing technology for blind and visually impaired employees
– Developing an auxiliary audio system for digital scales to output precise, non-visual measurements
– Adapting the native inkjet coder interface to be reprojected on an external monitor with color correction

Mechanical Engineering Intern – Regal Boats May 2023 – Aug 2023

– Preformed CFD analysis and validation on hull designs for running characteristics and excitation response
– Designed and installed propulsion, plumbing, and electrical systems for yacht manufacturing
– Measured boat vibration and running characteristics with DEWESOFT sensor and DAQ systems

PROJECTS

External Ventricular Drain Clearance Dec 2024 - Present

– Collaborating with a Washington University neuroscience research team to design and evaluate vibration-based catheter clearance systems
– Characterizing vibration profiles and quantifying their effects on fluid clearance efficiency in ventricular catheters

Computational Fluid Dynamics (CFD) for Microfluidic Flow Device Jan 2024 – May 2024

– Leveraged simulation results for shear stress and fluid dynamics to guide device design and validate fluid conditions, ensuring physiologically relevant cell culture experiments

Infrared Photogate for Anemia Confirmation- *2nd place Rice 360 Global Heath Challenge* Oct 2021 – May 2023

– Created a cost-effective method of automatic data collection for diagnosing Sickle Cell disease in developing nations
– Conducted a multiphase CFD study in OpenFOAM to evaluate alignment impact on droplet dynamics

EDUCATION

Mechanical Engineering | Saint Louis University

– **Master of Science (MS)**: 4.00/4.00 GPA [May 2026]; Lighthouse for the Blind Fellow
  Design and Evaluation of Wearable Haptic Devices for Non-Visual Guided Breathing
– **Bachelor of Science (BS)**: 3.99/4.00 GPA [Graduated May 2024]
  summa cum laude / Dean's List 2020-2024

Scholarships and Awards

Keynote speaker at SLU scholarship dinner | 2024, Patrick P. Lee Engineering Scholarship, Gerald E. Dreifke Engineering Scholarship, Roland Quest Memorial Scholarship, Dr. Bellur L. Nagabhushan Endowed Scholarship, Frank T. & Bernice P. Magiera Endowed Scholarship, SLU Vice President's Scholarship, Division 1 Baseball Scholarship

TECHNICAL SKILLS AND PROFICIENCIES

Mechatronic systems, computational fluid dynamics (CFD, Fluent, OpenFOAM, Orca3D), finite element analysis (FEA, ANSYS, ABAQUS), computer-aided design (CAD, SolidWorks, Rhino, Blender, NX/Unigraphics, Fusion 360), programming (Python, MATLAB, ROS2, Linux, C++/C), welding, rapid prototyping (3D printing, FDM, SLA), stress analysis, mechanoelectrical hardware and sensors (Arduino, ESP32, Raspberry Pi, etc.), marine propulsion and systems